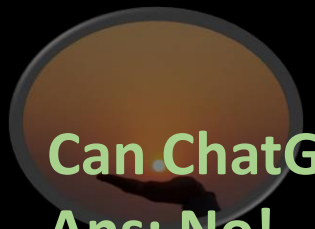
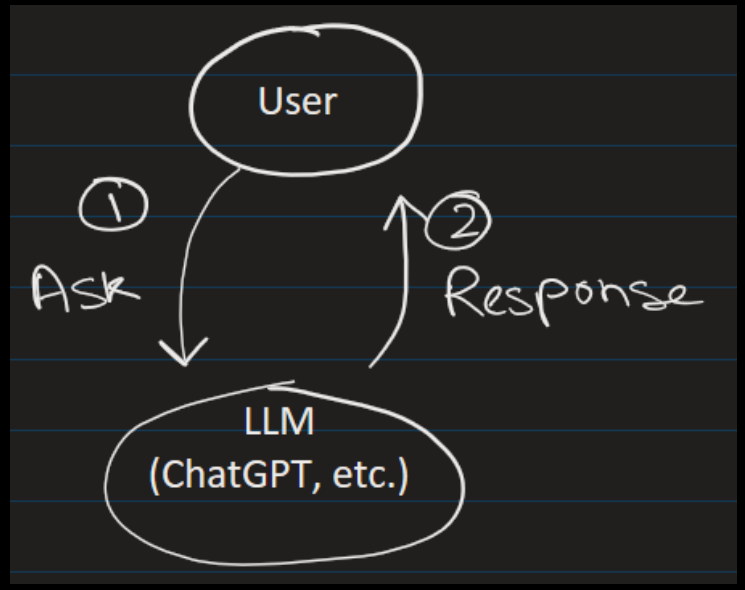
RAG
Retrieval-Augmented Generation



Can ChatGPT or any LLM answer questions about YOUR personal files?
Ans: No!

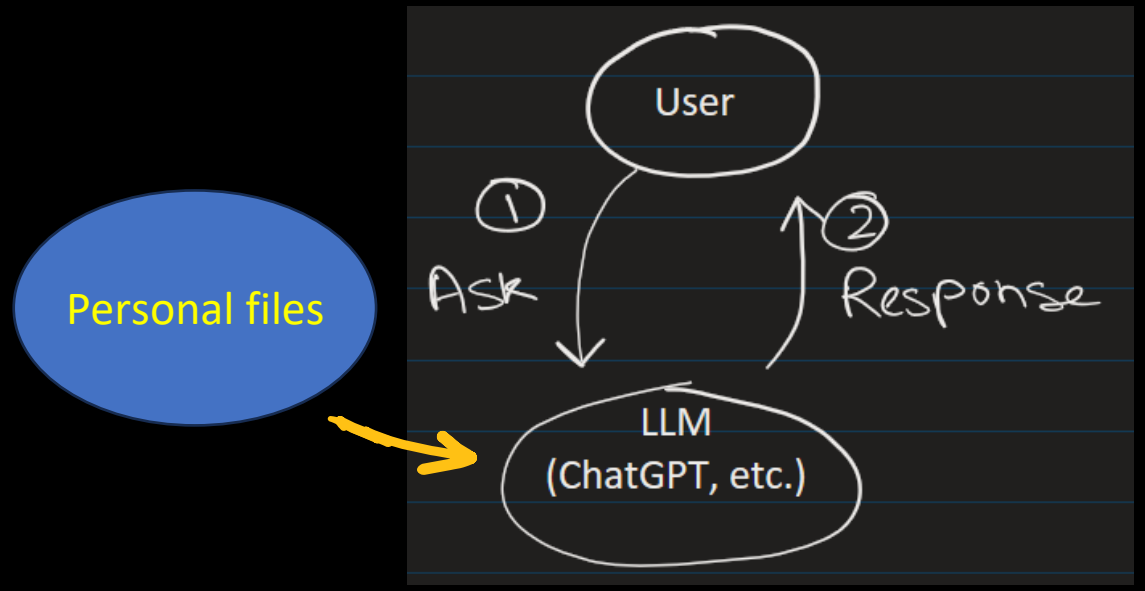
Personal files



RAG solves this types of problem.

What is RAG?

RAG (Retrieval-Augmented Generation) is a technique in **Generative AI** where a language model (like GPT) first **retrieves relevant information from an external source**, and then **uses that information to generate a better, more accurate response**.



Why do we need RAG?

Let's understand the problem first.

❌ **Problem with Normal LLMs**

Large Language Models (LLMs):

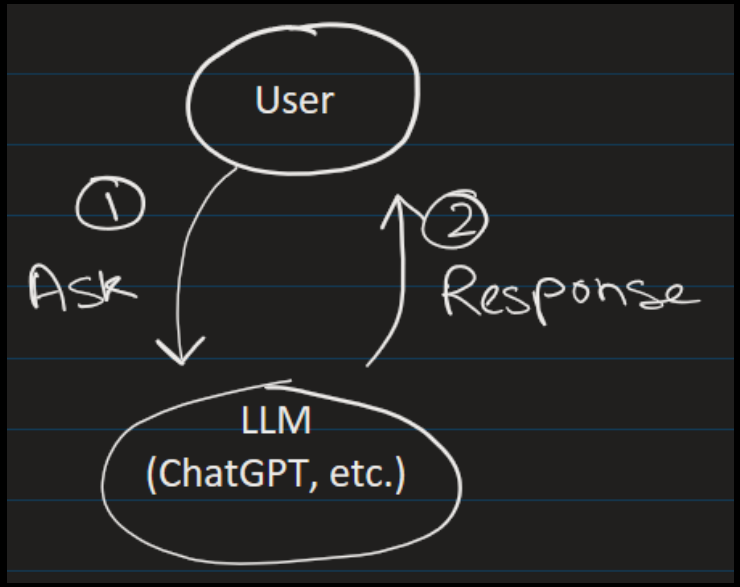
- Are trained on **fixed data (training data)**
- Do **not know recent updates**
- Can sometimes **hallucinate (make up answers)**

Example:

Ask: *"What is my company, Loxford Academy, policy?"*

A normal LLM will either:

- Guess ❌
- Give generic answer ❌

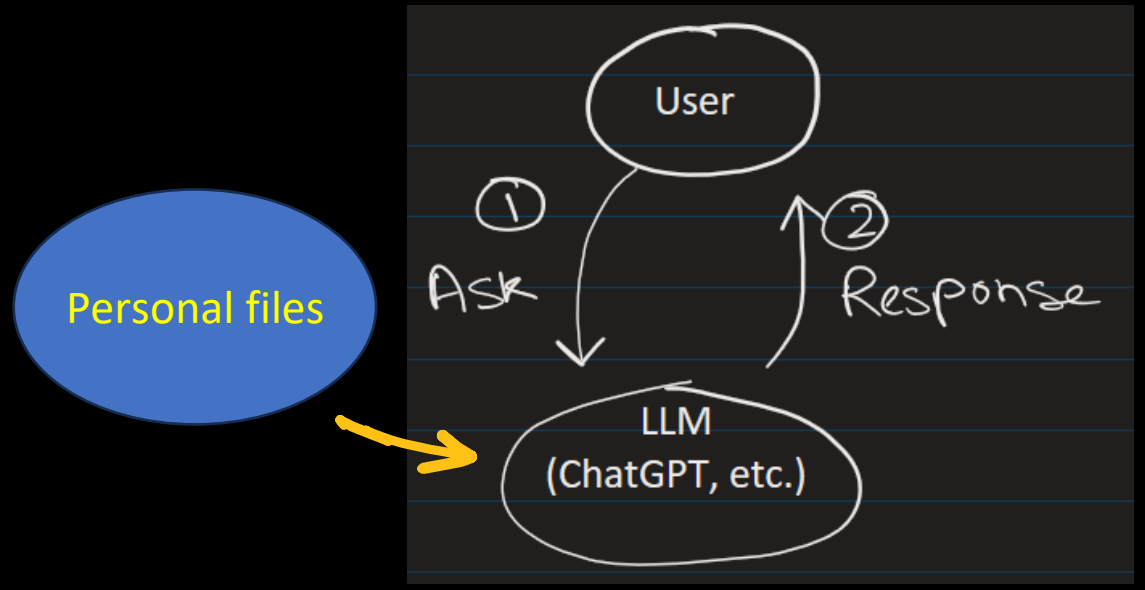


Normal Flow: Without RAG

How RAG solves this

RAG connects the model to **your own data source**:

- PDFs
- Databases, Websites, Company documents

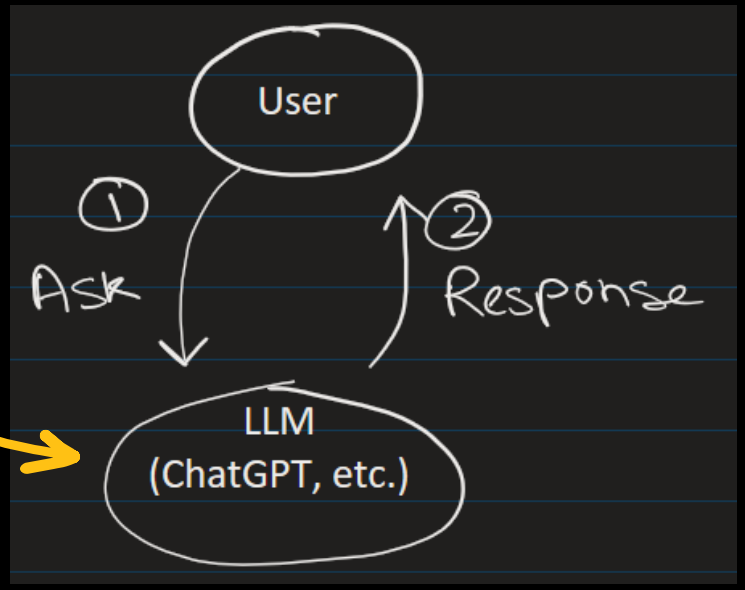


Now flow becomes:

1. User asks question
2. Search relevant documents and Extract useful information
3. Generate answer based on **your data source**

✓ Result:

- Accurate answers
- Up-to-date information



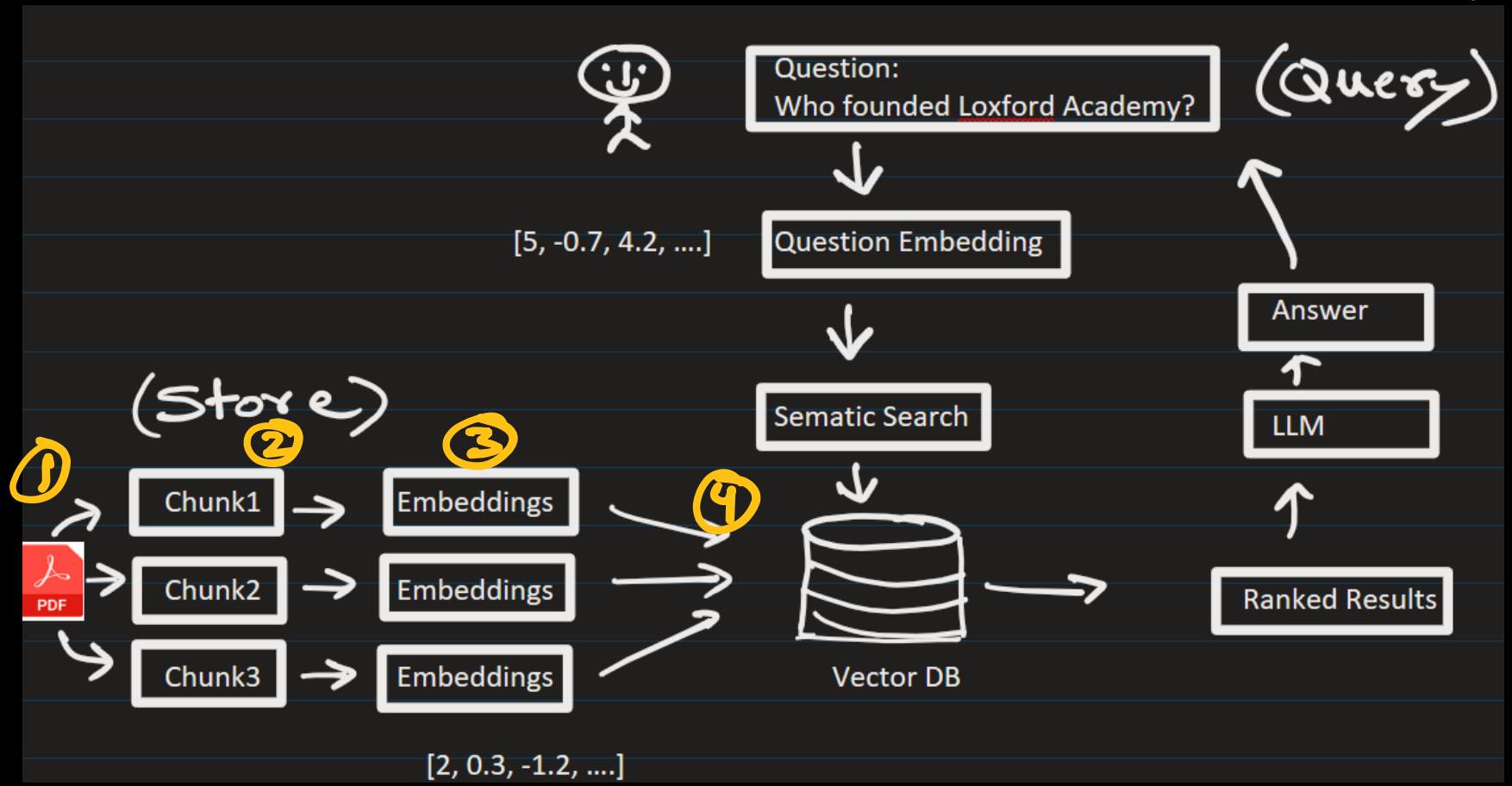
RAG Architecture/Process
Phase A (Populate The Vector DB)

1. Data Collection
Collect documents
(PDFs, text, etc.)

2. Chunking
Break data into small pieces
(chunks)

3. Embeddings
Convert text into vectors
using models

4. Storage
Store vectors in a database
called Vector DB
(e.g., Pinecone, FAISS, etc.)



Phase B (User Query)

1. Retrieval

When user asks a question (query):

- Convert query into vector
- Find similar data from Vector DB (AKA context)

Example of Query:

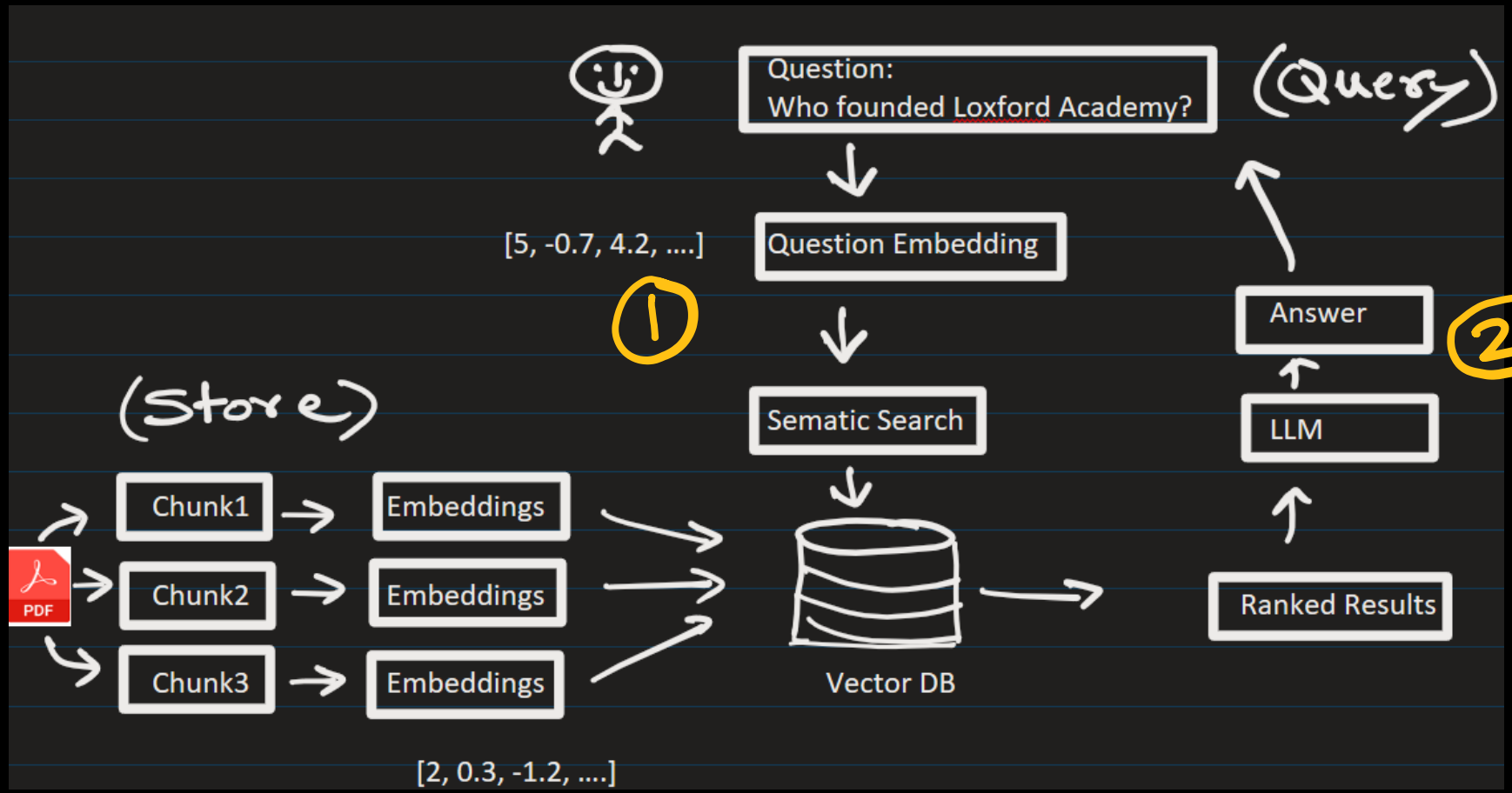
Who founded Loxford Academy ?

Ranked Results From VectorDB (Similar data):

1) Founder of Loxford Academy are living in Moscow

2) Ms. Lomax and Mr. Thomas founded Loxford Academy in 2018

3) One of the founders of Loxford Academy went to MIT.



Phase B (User Query)

2. Generation

- Pass (retrieved data + question + prompt) to LLM
- LLM generates final answer and sends it to user

PROMPT:

```
Answer ONLY from the context.

Context:
{context_from_VectorDB}

Question:
{user_query}
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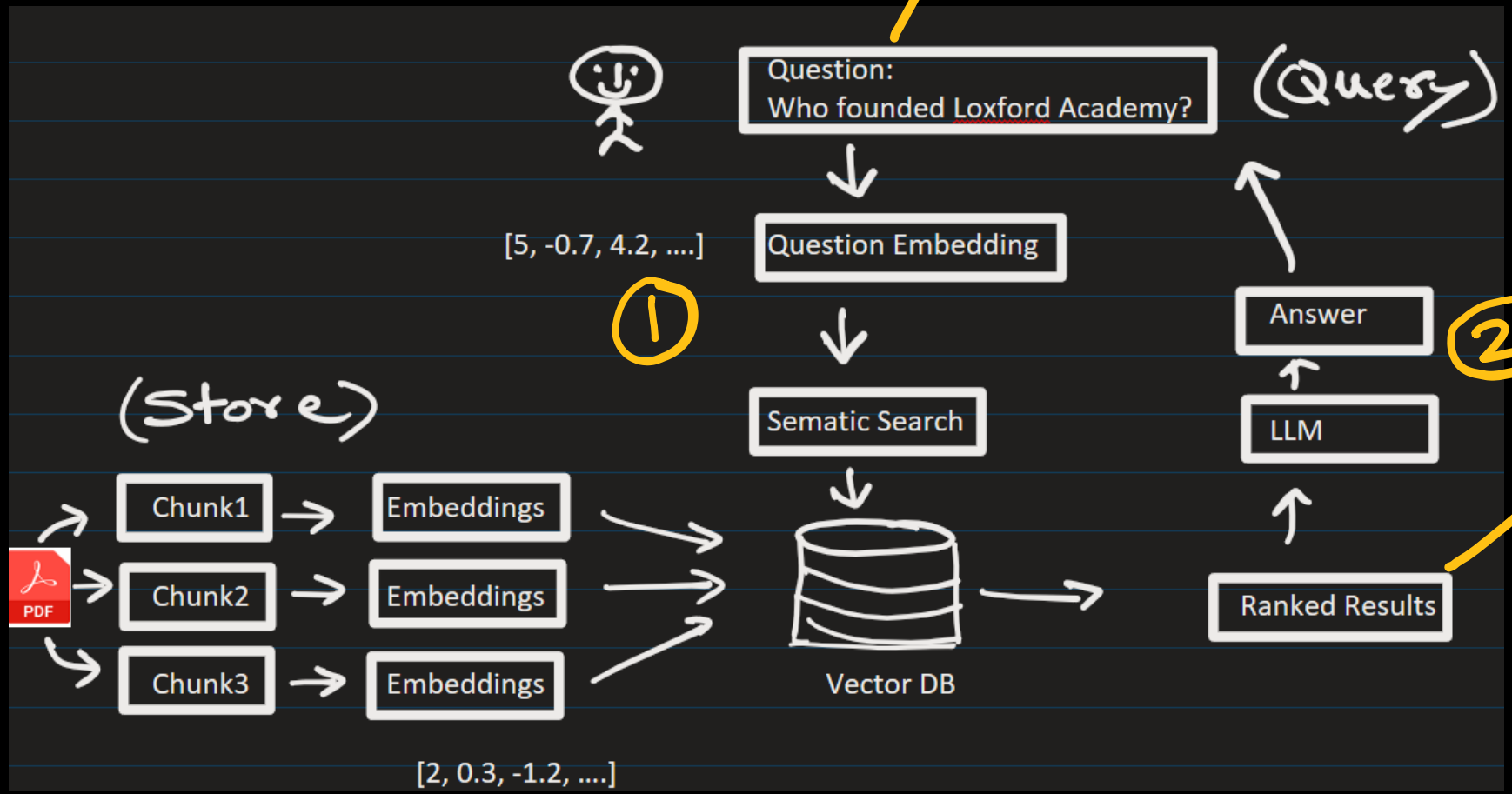


Phase B (User Query)

Prompt passed to LLM:

PROMPT:

Answer ONLY from the context.

Context:

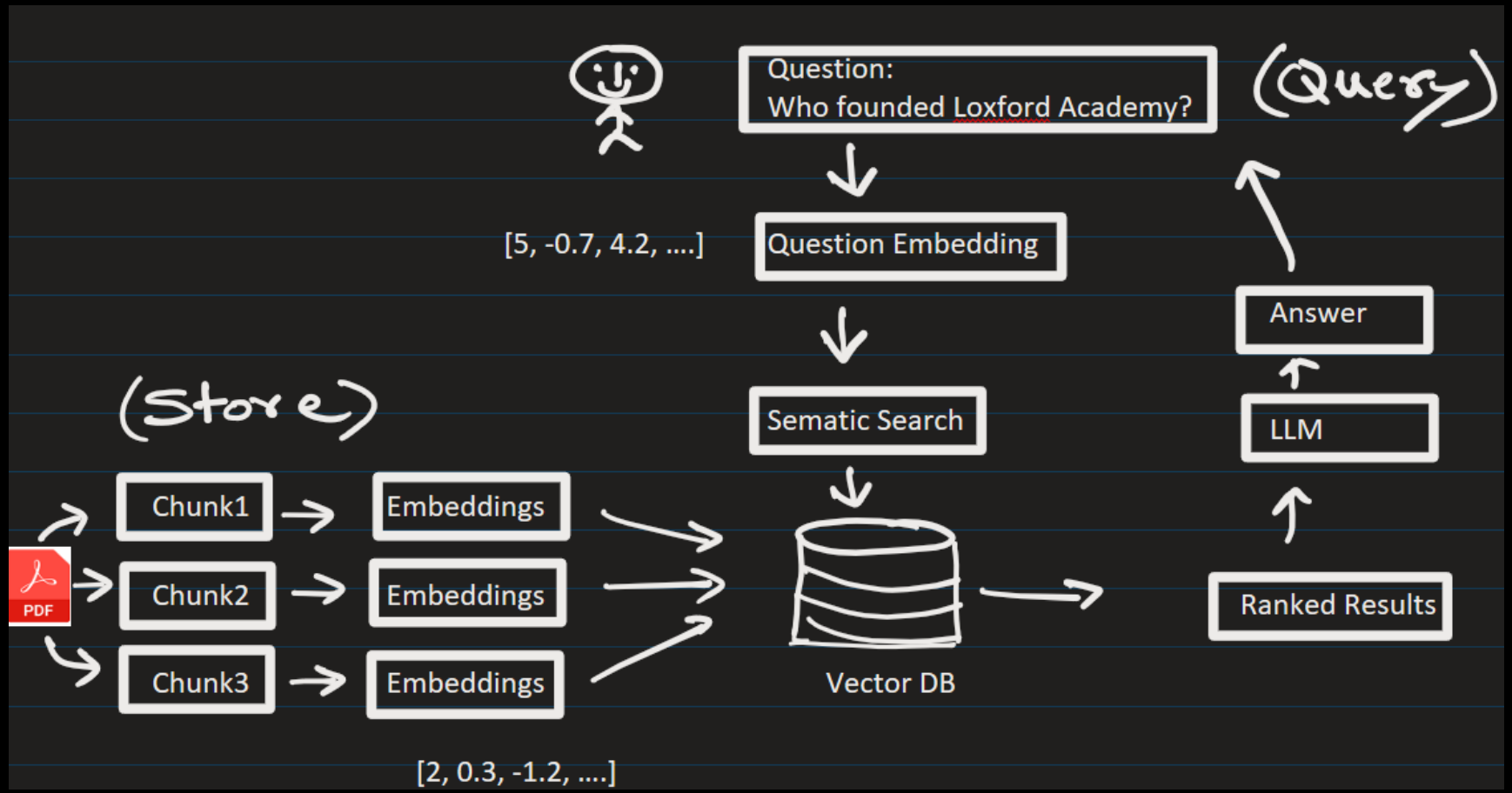
1) Founder of Loxford Academy are living in Moscow
2) Ms. Lomax and Mr. Thomas founded Loxford Academy in 2018
3) One of the founders of Loxford Academy went to MIT.

Question:

Who founded Loxford Academy ?

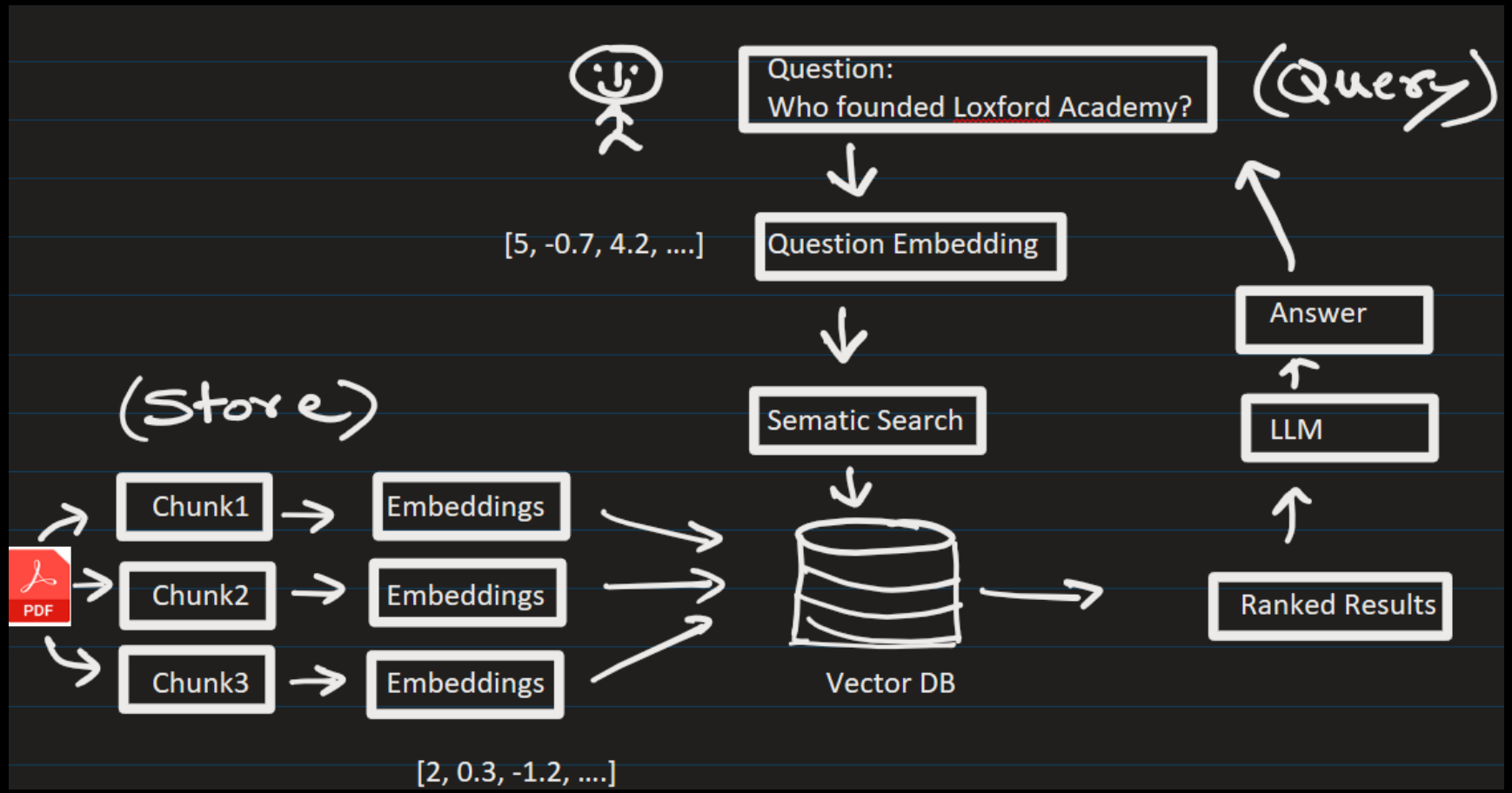
🔹 Key Benefits

✓ Reduces hallucination
✓ Keeps answers updated
✓ Works with private data



Where is RAG Used?

- Chatbots for companies
- Customer support systems
- Legal document search
- Medical record assistants
- Internal knowledge assistants



EXTRA

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